

Enhancing SIEM Systems Using Time Correlation

Senior Project 1

**Submitted in partial fulfillment of the requirements for the degree of
Bachelor of Engineering in Intelligent Information
Engineering in Intelligent Information Security Systems**

Prepared by:

Tala Almasalmeh

Supervised by:

Dr. Wassim Junidi

Academic Year: 2025–2026

Table of Contents

Contents

٨	Abstract
١٠	Chapter 1: Introduction
١١	Chapter Introduction 1.1
١١	Project Overview 1.2
١٢	Scientific Problem and Project Justification 1.3
١٣	Project Objectives 1.4
١٣	Research Questions 1.5
١٤	Importance of the Project 1.6
١٥	Related Studies Supporting the Project 1.7
١٦	Current State of the Art in the Project Area 1.8
١٦	Challenges Addressed by the Project 1.9
١٧	Practical Applications of the Project 1.10
١٧	General Scope of the Project 1.11
١٨	Thesis Structure 1.12
١٨	Chapter Conclusion 1.13
١٩	Chapter 2: Theoretical Framework
٢٠	2.1 Chapter Introduction
٢٠	2.2 Security Information and Event Management Systems
٢٠	2.3 Security Event Correlation in SIEM
٢١	2.4 Time Correlation
٢١	2.5 The Difference Between Pre-Processing and Pre-Correlation
٢١	2.6 Main Stages of Pre-Correlation
٢٢	2.7 Redundancy, Excessive Alerts, and Analyst Fatigue
٢٢	2.8 Source-Aware Processing
٢٢	2.9 The LANL Dataset as a Theoretical and Experimental Basis
٢٢	2.10 Baseline Correlation as a Reference Layer
٢٢	2.11 Context Preservation in Pre-Correlation
٢٣	2.13 Criteria for Evaluating Pre-Correlation Quality
٢٣	2.14 Chapter Conclusion

٢٤	Chapter 3: Work Environment and Methodology
٢٥	3.1 Chapter Introduction
٢٥	3.2 General Execution Environment
٢٦	3.3 Dataset and Selected Sources
٢٧	3.4 Raw Files and Actual Source Inputs
٢٨	3.5 Tools and Libraries Used
٢٩	3.6 Software Structure of the Project
٣٠	3.7 General Implementation Methodology
٣١	3.8 Methodology of the Main AUTH Path
٣٢	3.9 Methodology of the Exploratory Extensions
٣٢	3.10 Selection of Samples and Chunks
٣٣	3.11 Evaluation Logic
٣٣	3.12 Baseline Correlation Setup
٣٤	3.13 Redteam-Based Validation
٣٤	3.14 Output Management and Stored Results
٣٥	3.15 Chapter Conclusion
٣٦	Chapter 4: Proposed Solution
٣٧	4.1 Chapter Introduction
٣٧	4.2 General Architecture of the Proposed Solution
٣٨	4.3 Working Methodology of the Proposed Solution
٣٩	4.4 Main AUTH Path and Source-Aware Extensions
٤٠	4.5 Parsing and Initial Data Structuring
٤١	4.6 Normalization
٤١	4.7 Improved Filtering
٤١	4.8 Exact De-duplication
٤٢	4.9 Relaxed Temporal Aggregation
٤٢	4.10 Baseline Correlation Inside the Proposed Solution
٤٣	4.11 Chunk-Based Validation and Final Strategy Selection
٤٣	4.12 Exploratory Extensions for Other Sources
٤٤	4.13 Final Software Structure of the Solution
٤٥	4.14 Final Design Decisions and Their Justifications
٤٥	4.15 Overall Pipeline Logic

٤٦	4.16 Chapter Conclusion
٤٧	Chapter 5 : Comparisons and Results
٤٨	5.1 Chapter Introduction
٤٨	5.2 Method of Presenting the Results
٤٩	5.3 Results of the Main AUTH Path
٤٩	5.3.1 Stream Size Reduction
٤٩	5.3.2 Effect on Baseline Correlation
٥٠	5.3.3 Preservation of Attack-Related Context
٥٠	5.3.4 Summary of AUTH Results
٥١	5.3.5 Visual Summary of AUTH Results
٥٢	5.4 Chunk-Based Validation Results
٥٢	5.4.1 Validation Across 3 Chunks
٥٢	5.4.2 Validation Across 8 Redteam-Zone Chunks
٥٣	5.5 Results of the Exploratory Extensions
٥٣	5.5.1 PROC Results
٥٤	5.5.2 FLOWS Results
٥٤	5.5.3 DNS Results
٥٤	5.6 Aggregated Comparison Across Sources
٥٥	5.7 Discussion of Results
٥٦	5.7.1 Stronger Scientific Interpretation
٥٦	5.8 Comparison with Previous Studies
٥٧	5.8.1 Comparing the Project with Reference Studies
٥٨	5.9 Current Limitations of the Results
٥٨	5.10 Chapter Conclusion
٥٩	Conclusion
٦١	References

List of Tables

١٥	Table 1 : Related Studies
٥٠	Table 2 : Main AUTH Path Results Summary
٥٥	Table 3 : Exploratory Extensions and Cross-Source View
٥٧	Table 4 : Comparison with Similar Reference Studies

List of Figures

၃၈	Figure 1 : General Proposed Solution
၃၉	Figure 2 : Proposed Solution Implementation Mechanism
၄၀	Figure 3 : Main AUTH Path and Source-Aware Extensions
၄၄	Figure 4 : Final Modular Software Structure
	Figure 5 : compares AUTH event counts before and after processing. After filtering +
၅၁	relaxed temporal aggregation
	Figure 6 : : Main AUTH Path Results showing the change in correlation outputs from
၅၁	360 to 337. Reduction of 23 outputs(6,3,9%)
၅၃	Figure 7 : AUTH Validation Across Redteam-Zone Chunks

List of Abbreviations and Technical Terms

Abbreviation	Meaning
SIEM	Security Information and Event Management
IDS	Intrusion Detection System
AUTH	Authentication Data Source
PROC	Process Event Data Source
DNS	Domain Name System Records
FLows	Network Flow Records
LANL	Los Alamos National Laboratory
CSV	Comma-Separated Values

Abstract

This project addresses the topic of Enhancing SIEM systems using time correlation by designing and implementing a temporal pre-correlation layer before baseline correlation in order to improve the security event stream before downstream analysis. The problem arises from the fact that modern SIEM systems ingest large volumes of heterogeneous and time-stamped alerts and logs, and these raw streams often contain redundancy, noise, and fragmented evidence, which reduce analytical clarity and increase the downstream burden on correlation outputs. Therefore, the project aims to reduce stream volume before correlation while preserving attack-related context as much as possible.

The project was implemented as a Python prototype using the LANL Cyber Security Dataset, with AUTH as the main path and PROC, FLOWS, and DNS as exploratory extensions under a source-aware pre-correlation framework. The proposed solution included parsing, normalization, improved filtering, exact de-duplication, relaxed temporal aggregation, baseline correlation, and before/after evaluation.

The results of the main AUTH path showed that the event stream size decreased from 10000 events to 9439, while correlation outputs decreased from 360 to 337, representing a 6.39% reduction. At the same time, redteam-related context remained close to its original level, changing only from 73.15% to 72.67%. The exploratory extensions also supported the central project idea and showed that the framework can remain unified, but implementation strategies must still depend on the nature of each source.

The results indicate that improving the stream before correlation is feasible, measurable, and scientifically justified, and that time in pre-correlation can serve as an effective analytical factor for reducing noise and improving event representation before correlation.

Keywords:

SIEM, Time Correlation, Pre-Correlation, Temporal Aggregation, AUTH, LANL Dataset, Baseline Correlation, Source-Aware Processing

الملخص

يتناول هذا المشروع موضوع Enhancing SIEM systems using time correlation من خلال تصميم وتنفيذ طبقة temporal pre-correlation تسبق baseline correlation، بهدف تحسين stream الأمنية قبل التحليل اللاحق. تنطلق المشكلة من أن أنظمة SIEM الحديثة تستقبل كميات كبيرة من الأحداث والتنبيهات غير المتجانسة والموسومة زمنيًا، وأن هذه البيانات الخام غالبًا ما تحتوي على redundancy و noise و fragmented evidence، مما يضعف وضوح التحليل ويزيد العبء على correlation outputs downstream. لذلك، يهدف المشروع إلى تقليل حجم stream قبل correlation، مع الحفاظ قدر الإمكان على attack-related context.

نُفذ المشروع بوصفه Python prototype باستخدام LANL Cyber Security Dataset، مع اعتماد AUTH كمسار رئيسي و PROC و FLOWS و DNS كتوسعات استكشافية ضمن إطار source-aware pre-correlation. وتضمن الحل المقترح مراحل parsing و normalization و improved filtering و exact de-duplication و relaxed temporal aggregation ثم baseline correlation و before/after evaluation.

أظهرت نتائج المسار الرئيسي AUTH انخفاض حجم stream من ١٠٠٠٠ حدث إلى ٩٤٣٩، وانخفاض correlation outputs من ٣٦٠ إلى ٣٣٧، أي بنسبة ٦.٣٩٪، مع بقاء redteam-related context قريبًا من مستواه الأصلي، إذ تغير من ٧٣.١٥٪ إلى ٧٢.٦٧٪ فقط. كما دعمت التوسعات الاستكشافية الفكرة المركزية للمشروع، وأظهرت أن framework يمكن أن تكون موحدة، لكن الاستراتيجيات التطبيقية يجب أن تبقى مرتبطة بطبيعة المصدر.

تشير النتائج إلى أن تحسين stream قبل correlation ممكن، قابل للقياس، ومبرر علميًا، وأن استخدام الزمن داخل pre-correlation يمكن أن يكون عاملاً تحليليًا فعالاً في تقليل الضوضاء وتحسين تمثيل الأحداث قبل correlation.

Chapter 1: Introduction

1.1 Chapter Introduction

Security Information and Event Management (SIEM) systems have become a central component of modern digital environments because they collect and process large volumes of security logs and alerts from multiple sources. However, the main practical challenge is not only collecting events, but also improving the quality of the event stream before it reaches downstream correlation. Large SIEM streams often contain redundancy, noise, and fragmented evidence, which reduce analytical clarity and increase the burden on later analysis stages [1], [2], [3]

Recent research therefore supports the idea that stream improvement before correlation is an important part of modern security analytics rather than a secondary cleanup step [1], [4].

This project addresses that challenge by introducing a temporal pre-correlation layer before baseline correlation. Its purpose is to compact the raw event stream, reduce low-value repetition, and preserve attack-related context as much as possible. This direction is consistent with the project's final scope and presentation materials, which define the work as a source-aware pre-correlation prototype with AUTH as the main path and PROC, FLOWS, and DNS as exploratory extensions.

1.2 Project Overview

This project focuses on improving event analysis in SIEM systems by designing a temporal pre-correlation layer that operates before baseline correlation. The main idea is that raw security events do not always reach correlation in a suitable form. They may be repetitive, low-value, or temporally close in a way that makes reduction or re-representation necessary before later analysis. [1], [2]

In practical terms, the project implements a prototype based on the LANL Cyber Security Dataset, with AUTH as the main implementation path and PROC, FLOWS, and DNS as exploratory extensions. The project adopts a source-aware design, which means that the overall pre-correlation concept is unified, while the actual reduction strategy varies depending on the structure of the source [2], [4], [5].

The selected dataset, the AUTH main path, and the exploratory role of the other sources are all part of the project's defined methodology and implementation scope.

1.3 Scientific Problem and Project Justification

The scientific problem addressed in this project is that modern SIEM platforms ingest large volumes of heterogeneous, time-stamped alerts and logs. When these raw events are sent directly to correlation, redundancy, alert flooding, and temporally fragmented evidence reduce analytical clarity and weaken the usefulness of downstream correlation .[1], [2], [3]

Therefore, the need is not only for more correlation, but for a time-aware pre-correlation stage that groups temporally related events, removes duplicates and noise, and delivers a cleaner stream to baseline correlation [1], [2].

The project is justified by several connected reasons:

- **First, there is a need to reduce the event stream before correlation instead of focusing only on improving correlation itself. [1], [2]**
- **Second, time should be used as an analytical dimension in grouping and aggregation, not only as a descriptive field. [2], [7], [8]**
- **Third, reduction must be evaluated carefully to ensure that attack-related context is not lost. [5], [6]**
- **Fourth, the framework should remain source-aware, since a single strategy is unlikely to fit all event sources.[2], [4]**

These justifications are also reflected in the project's own problem statement and goals, which define the work as a time-aware pre-correlation layer rather than a replacement for SIEM correlation.

1.4 Project Objectives

The general objective of this project is to improve SIEM analysis by introducing a temporal pre-correlation stage before baseline correlation, so that the resulting event stream becomes more compact, more meaningful, and easier to correlate [1], [2], [3].

The specific objectives are as follows:

- Parse raw security events into a structured format suitable for later analysis [1], [4].
- Normalize temporal and semantic fields required for comparison and grouping [1], [2].
- Filter low-value events and remove direct duplication before correlation [1], [4].
- Aggregate temporally related events into higher-value grouped events [1], [2].
- Measure the effect of pre-correlation on baseline correlation outputs [1], [2].
- Evaluate whether attack-related context is preserved after reduction [5], [6].
- Test the adaptability of the idea across multiple data sources [2], [4], [5].

These objectives are identical in spirit to the project objectives stated in the uploaded seminar materials.

1.5 Research Questions

This project seeks to answer the following research questions:

1. Can pre-correlation techniques reduce the size of the security event stream before baseline correlation? [1], [2]
2. Does this reduction decrease downstream correlation outputs? [1], [2], [4]
3. Can reduction be achieved while preserving attack-related context? [5], [6]
4. Can one strategy work for all data sources, or does each source require a different approach? [2], [4], [5]
5. How can time be used to transform raw events into higher-value analytical units? [2], [7], [8]

1.6 Importance of the Project

The importance of this project can be explained from three related perspectives.

- **Scientific importance:** The project addresses a stage that is often less emphasized than detection or correlation itself, namely pre-correlation with a temporal focus [1], [2].
- **Practical importance:** Reducing redundancy and noise before correlation can improve analytical clarity, lower downstream burden, and reduce analyst overload [3], [5], [6].
- **Methodological importance:** The project proposes a unified framework, but insists that implementation decisions must remain source-aware, which makes the approach more realistic than blind uniform processing [2], [4], [5].

This importance is also visible in the project's own executive overview, which states that the goal is to improve the stream before correlation and to preserve attack-related context while reducing downstream load.

1.7 Related Studies Supporting the Project

The project was supported by a set of recent studies from complementary angles:

Table 1 : Related Studies

Paper Title	Researcher(s)	Year	Achievement	Relevance to My Project
Revolutionizing SIEM Security: An Innovative Correlation Engine Design for Multi-Layered Attack Detection	Muhammad Sheeraz et al.	2024	Designed a SIEM correlation engine with explicit event filtering, aggregation, and de-duplication stages for layered attack detection.	Directly supported the decision to include filtering, de-duplication, and aggregation before baseline correlation.
A Hierarchical Security Event Correlation Model for Real-Time Threat Detection and Response	Herbert Maosa et al.	2024	Built a hierarchical model that aggregates events into higher-level units before later correlation and response.	This was the strongest support for choosing temporal aggregation before baseline correlation.
Improving SIEM alert metadata aggregation with a novel kill-chain based classification model	Blake D. Bryant; Hossein Saiedian	2020	Improved SIEM alert metadata aggregation using a kill-chain based classification model.	Supported the decision to preserve meaningful event attributes after aggregation so that grouped events remain analytically useful.
HTTPS Event-Flow Correlation: Improving Situational Awareness in Encrypted Web Traffic	Stanislav Špaček et al.	2022	Correlated host and network data using common features and a correlation time-window, while studying preprocessing effects on correlation quality.	Supported the decision to keep time and event ordering central to the project design.
Processing of IDS alerts in multi-step attacks	Tomáš Bajtoš et al.	2024	Presented a software flow including alert preprocessing, filtration, and aggregation for multi-step attack analysis.	Strengthened the idea that raw alerts should be reduced and consolidated before downstream correlation.
Intrusion Alert Analysis Method for Power Information Communication Networks Based on Data Processing Units	Rui Zhang et al.	2025	Addressed alert redundancy and amplification by transforming alerts into more manageable analysis units.	Supported the problem statement about alert noise, redundancy, and the need for pre-correlation reduction.
Near real-time intrusion alert aggregation using concept-learning	Gordon Werner; Shanchieh Jay Yang	2021	Proposed near real-time alert aggregation through learned temporal concepts over streaming alerts.	Supported the decision to keep temporal grouping close to the stream rather than as an offline cleanup step only.
CLEAR-ROAD: Extraction of Temporally Co-occurring yet Rare Critical Alerts	Gordon Werner; Shanchieh Jay Yang	2021	Focused on extracting rare but temporally co-occurring critical alerts from streaming IDS data.	Directly reinforced the use of temporal co-occurrence as a grouping signal in pre-correlation.
Finding Harmony in the Noise: Blending Security Alerts for Attack Detection	Tom-Martijn Roelofs et al.	2024	Blended multiple security alerts to reduce the number that require analyst investigation while keeping attack detection effective.	Supported the practical value of the project for analysts by reducing alert volume without discarding attack structure.
Efficient Post-Processing of Intrusion Detection Alerts Using Data Mining and Clustering	Kalu Kumarasinghe et al.	2025	Applied post-processing, data mining, and clustering to improve the quality of IDS alert streams.	Helped position the project as a focused pre-/post-correlation processing layer rather than a full SIEM replacement.
Breaking Alert Fatigue: AI-Assisted SIEM Framework for Effective Incident Response	Tao Ban et al.	2023	Formulated alert fatigue as a major SIEM challenge and separated filtering from correlation concerns in incident response workflows.	Supported the scientific problem statement: raw high-volume alerts reduce analyst efficiency and motivate a noise-reduction stage before correlation.

1.8 Current State of the Art in the Project Area

Recent research shows several clear trends in this area.

- First, there is a growing move from treating each alert or event as an isolated record toward building higher-level grouped units through aggregation or hierarchical correlation [1], [2].
- Second, recent work increasingly emphasizes noise reduction and alert-fatigue mitigation before downstream analysis [3], [5], [6].
- Third, time is being used not only in later analysis but also in the grouping process itself through temporal windows and temporal concepts [2], [7], [8]
- Fourth, more convincing solutions increasingly respect the structure of each source instead of assuming that one strategy can fit every type of data [2], [4], [6].

These same themes are summarized in the project's related-work slides as security event correlation, correlation engine design, hierarchical correlation, and alert fatigue.

1.9 Challenges Addressed by the Project

This project addresses several scientific and practical challenges:

- the high volume and structural diversity of security events [1], [2]
- the presence of redundancy and low-value noise in raw streams [3], [6]
- the temporal fragmentation of attack-related evidence [2], [7]
- the difficulty of using one reduction strategy for all sources [2], [4]
- and the need to balance stream reduction with context preservation [1], [5]

These challenges explain why pre-correlation should not be treated as a minor cleanup stage, but as an analytical layer that affects the quality of later security analysis.

1.10 Practical Applications of the Project

From a practical perspective, the proposed pre-correlation layer can be viewed as a stage that may be integrated before correlation in SIEM environments. Such a layer can help to:

- reduce the size of the event stream before later analysis [1], [3]
- lower the burden on correlation engines [1], [2],
- improve the analyst's ability to follow evidence more clearly [3], [5],
- reduce the number of low-value events reaching advanced analysis stages [1], [4],
- and support the construction of more meaningful behavioral groups instead of relying only on isolated raw rows [2], [7].

The source-aware nature of the framework also makes it more realistic, since enterprise security environments rely on multiple heterogeneous data sources rather than a single homogeneous stream [2], [4]

1.11 General Scope of the Project

This work is a practical prototype focused on building and testing an interpretable pre-correlation pipeline. It does not aim to build a full production SIEM or a commercial-grade correlation engine, and it does not rely on machine learning in its current version. Instead, it focuses on one clear main path based on authentication data, with exploratory extensions for additional sources and structured before/after evaluation. This scope is consistent with the project implementation itself, where AUTH is the main path and the other sources are treated as exploratory extensions.

1.12 Thesis Structure

This report is organized into several connected chapters, each building upon the previous one in order to provide a complete presentation of the research problem, theoretical basis, practical implementation, and obtained results.

- **Chapter 1: Introduction:** This chapter presents the scientific background, the research problem, the project objectives, the research questions, the importance of the work, the related studies, current developments in the field, the main challenges, and the practical applications.
- **Chapter 2: Theoretical Framework:** This chapter discusses the fundamental concepts and theories related to the project.
- **Chapter 3: Work Environment and Methodology:** This chapter explains the selected dataset, the software environment, the tools and libraries used, and the implementation methodology.
- **Chapter 4: Proposed Solution:** This chapter presents the design of the implemented pre-correlation pipeline and the final design decisions.
- **Chapter 5: Results and Discussion:** This chapter presents the practical results and discusses their meaning, limitations, and implications.
- **Conclusion:** This final part summarizes the project and presents the final recommendations

1.13 Chapter Conclusion

This chapter presented the scientific and practical foundation of the project and showed that the challenge in SIEM environments is not limited to building stronger correlation alone, but also includes improving the quality of the event stream before correlation. It also showed that the project is grounded in a recent research direction concerned with reducing redundancy and noise, exploiting temporal relations in grouping, preserving attack-related context, and adopting source-aware strategies instead of blind uniform processing. Based on this introduction, the next chapter moves to the theoretical framework that supports the proposed solution. [1], [2], [3], [5]

Chapter 2: Theoretical Framework

2.1 Chapter Introduction

This chapter presents the theoretical foundation on which the project is built. It explains the concepts related to Security Information and Event Management (SIEM) systems, security event correlation, the role of time in analysis, the distinction between general pre-processing and pre-correlation, and the main stages involved in this type of processing. The importance of this chapter lies in linking the scientific idea of the project to its practical implementation, so that stages such as parsing, normalization, filtering, and aggregation are not treated as isolated technical steps, but as direct extensions of theoretical concepts used in recent literature to improve security event analysis [9].

2.2 Security Information and Event Management Systems

Security Information and Event Management (SIEM) systems are among the central tools used in modern security environments because they collect logs and events from multiple sources, then organize, analyze, and correlate them in order to support threat detection and response. The value of SIEM lies in its ability to combine heterogeneous data coming from different layers of the digital environment, such as authentication logs, system events, network flows, and DNS activity, into a unified analytical environment [9].

2.3 Security Event Correlation in SIEM

Security event correlation refers to the process of linking multiple events, alerts, or records in order to extract a higher analytical meaning than that provided by a single event alone. In many cases, one event is not sufficient to indicate an attack or suspicious behavior, but a group of related events can reveal a more meaningful sequence or pattern. Therefore, correlation is a core stage inside SIEM because it transforms multiple partial observations into a more coherent analytical picture [10].

2.4 Time Correlation

Time is one of the most important dimensions in security event analysis because events are not understood only through their type or source, but also through their order, temporal proximity, and relation to a broader time context. Recent literature supports this direction through concepts such as temporal windows, temporal concepts, and temporally co-occurring alerts. In this project, this concept directly supports temporal aggregation, because grouping is not based only on structural similarity, but also on proximity within a selected time window [10], [15], [16].

2.5 The Difference Between Pre-Processing and Pre-Correlation

Although pre-processing and pre-correlation may appear similar, the distinction between them is important in this project. General pre-processing includes any step intended to clean, transform, or prepare data for analysis, such as parsing, normalization, or handling missing values. Pre-correlation, on the other hand, is specifically intended to improve the event stream before correlation [12].

2.6 Main Stages of Pre-Correlation

The project adopts a set of connected stages that together form a pre-correlation pipeline. Although the importance of each stage varies depending on the source, the overall framework remains the same: improving the event stream before baseline correlation. These stages are consistent with recent literature on alert filtration, aggregation, and hierarchical grouping [9], [10], [12]:

- **Parsing:** reading raw data and converting it into a structured form.
- **Normalization:** unifying temporal and semantic fields for comparison.
- **Filtering:** removing low-value or high-noise events.
- **De-duplication:** removing direct repetitions.
- **Temporal Aggregation:** merging similar and temporally close events.
- **Evaluation:** measuring before/after behavior in terms of stream size and context.

2.7 Redundancy, Excessive Alerts, and Analyst Fatigue

One of the most important practical problems in modern SIEM environments is alert fatigue, meaning the analyst's overload caused by excessive alerts and dense event streams. Recent studies discuss this problem explicitly and show that noise and alert flooding affect not only system performance, but also the analyst's ability to make timely and accurate decisions [11], [13], [14].

2.8 Source-Aware Processing

Different security sources should not be processed in exactly the same way. Authentication logs differ structurally and semantically from process events or network flows. Therefore, even if the general pre-correlation framework is unified, the rules for filtering, aggregation, and evaluation should remain source-aware [10], [12].

2.9 The LANL Dataset as a Theoretical and Experimental Basis

The LANL Comprehensive, Multi-Source Cyber-Security Events dataset is suitable because it provides a multi-source and time-stamped environment. This makes it appropriate for studying the relationship between event structure, time, and attack-related context. It includes multiple types of enterprise-style events, making it suitable for multi-source security analysis [17].

2.10 Baseline Correlation as a Reference Layer

This project uses the concept of baseline correlation as a reference layer for comparing the event stream before and after pre-correlation. It provides a reference mechanism that makes it possible to compare the raw stream with the processed stream and to demonstrate that reduction is not merely a formal decrease in row count [10].

2.11 Context Preservation in Pre-Correlation

Any reduction process before correlation may appear successful if it decreases the number of rows. However, that success becomes misleading if it causes the loss of attack-related context. The goal is to reduce redundancy while preserving the behavioral structure and analytical evidence necessary for understanding the attack [13].

2.12 Temporal Aggregation Strategies

Temporal aggregation strategies rely on two main principles: similarity between events in selected fields, and temporal proximity within a defined time window. Ignoring time may lead either to merging events that do not belong to the same behavior or to leaving repetitive events separate when they should be grouped [9], [10], [12].

2.13 Criteria for Evaluating Pre-Correlation Quality

The quality of pre-correlation is evaluated through multiple criteria, including: the amount of event reduction achieved, the change in correlation outputs, and the degree of preservation of attack-related context [9], [10], [13].

2.14 Chapter Conclusion

This chapter presented the main theoretical concepts on which the project is based. It showed that improving the event stream before correlation is supported by foundations in recent literature. Based on this, the next chapter moves to the work environment and methodology.

Chapter 3: Work Environment and Methodology

3.1 Chapter Introduction

This chapter presents the practical environment in which the project was implemented and the methodology used to transform the theoretical idea into a measurable prototype. Its importance lies in explaining how the project moved from the scientific problem and the concepts of pre-correlation to actual implementation on LANL data, while clarifying the tools and libraries used, the nature of the selected data, the evaluation logic, and the way the project was organized programmatically. The chapter also explains that the project was not implemented as a full operational SIEM platform, but as a practical framework for testing the effect of pre-correlation before baseline correlation. Therefore, the discussion of the work environment and methodology does not only describe the tools, but also explains the logic of data selection, experiment preparation, evaluation design, and output interpretation.

3.2 General Execution Environment

The project was implemented in a local execution environment rather than a distributed or production environment, because its main objective was to build an interpretable and measurable prototype, not a full operational SIEM system. This choice was appropriate for the nature of the work because it allowed flexible data inspection, repeated experimentation, and direct comparison between alternative strategies, while keeping the implementation process clear and easy to review. Recent work on correlation-oriented pipelines and alert-processing frameworks often begins in prototype-style environments before moving toward broader operational settings [18], [19].

From a practical perspective, the project relied on a Python-based local environment, which made it possible to execute parsing, normalization, filtering, aggregation, and evaluation in an organized and iterative way. This was methodologically appropriate because the project does not claim real-time production deployment. Instead, it focuses on measuring the effect of pre-correlation before baseline correlation. In this sense, the execution environment was not only a technical choice, but part of the project design itself, since it supported reproducible and interpretable experimentation.

3.3 Dataset and Selected Sources

The project relied on the LANL Cyber Security Dataset as its main data environment. This dataset was selected because it provides multi-source, time-stamped enterprise-style security events, which makes it suitable for a project focused on temporal pre-correlation. The dataset includes multiple event sources and supports time-aware analysis in a context that is closer to enterprise security environments than single-source benchmarks [23].

In practice, the sources were not used in the same way. Instead, they were distributed according to a source-aware logic as follows:

- **AUTH:** the main source of the project, used to build the complete end-to-end pipeline.
- **PROC:** an exploratory extension focused on process events.
- **FLows:** an exploratory network-oriented extension.
- **DNS:** an exploratory communication-oriented extension.
- **Redteam:** a validation reference rather than a processing stream.

This distribution reflects the methodological logic of the project, where one source is developed as the main implementation path while the others are used to test how far the pre-correlation idea can be extended across different event structures.

3.4 Raw Files and Actual Source Inputs

The project relied on a set of raw files derived from the LANL dataset, with each file representing a specific type of security event. Each source played a different role depending on the type of event and the semantic structure it carried.

The AUTH files represented authentication events and formed the main source of the project because they included user, computer, authentication-type, and result-related information, which made them the most suitable source for building a complete pipeline that could measure stream reduction, correlation impact, and context preservation.

The PROC files represented process lifecycle events.

The FLOWS files represented network flow records.

The DNS files represented communication records between hosts.

The Redteam was used as a validation source rather than a processing stream [23].

From a methodological perspective, these files were not treated as equal inputs. AUTH was the main implementation path, while PROC, FLOWS, and DNS were used to test whether the pre-correlation concept could be extended to event structures with different semantics and operational meaning.

3.5 Tools and Libraries Used

The project was implemented using Python as the main programming language. Pandas was used for DataFrame construction, file loading, and row and column manipulation. Visual Studio Code was used as the main development environment, and Jupyter Notebook was used during the exploratory and analytical stages. Git and GitHub were used for code tracking and version control.

These tools were appropriate for several reasons:

- they support structured data processing efficiently,
- they allow rapid prototyping and iterative modification,
- they make it easy to save summaries and comparisons as CSV and TXT files,
- and they support modular project organization.

Therefore, the selected tools were not arbitrary. They were directly suited to a project that processes relatively large event streams in a local prototype environment.

3.6 Software Structure of the Project

The implementation gradually evolved from separate scripts into a clearer modular structure. This final structure included major elements such as:

- shared functions for loading, evaluation, and aggregation,
- separate strategy files for each source,
- a registry defining the properties of each source,
- a unified runner for executing the pipeline across sources,
- and dedicated files for exporting the final results.

This structure served the project in several ways:

- it made the implementation easier to follow.
- It supported extension to multiple sources.
- it simplified repeated execution.
- it improved traceability between code and results.
- it helped transform the work from isolated experiments into a more coherent framework.

3.7 General Implementation Methodology

The general methodology followed a sequence of stages that starts with raw data loading and ends with before-and-after comparison against baseline correlation. This sequence can be summarized in the following stages:

1. Parsing
2. Normalization
3. Filtering
4. De-duplication
5. Temporal Aggregation
6. Baseline Correlation
7. Evaluation

In practical terms, the methodology was applied through the following steps:

- selecting an appropriate source for experimentation, with AUTH as the main path
- reading raw data and converting it into structured form
- applying normalization when required, especially in AUTH
- applying a reduction strategy appropriate to each source
- comparing before-and-after behavior in terms of stream size and downstream effect
- repeating the tests across multiple chunks when possible.

This sequence is not merely a technical order. It is the practical logic that translates the theoretical concepts of Chapter Two into executable stages [18], [19], [21].

3.8 Methodology of the Main AUTH Path

AUTH is the main path of the project, and it is the only source that developed into a complete end-to-end pipeline including parsing, normalization, filtering, exact de-duplication, relaxed temporal aggregation, baseline correlation, and before-and-after evaluation.

The main design decisions in this path were:

- **improved filtering**
- **exact de-duplication**
- **relaxed temporal aggregation**
- **grouping by `src_user`, `dst_computer`, and `event_type`**
- **a time window of 5.**

AUTH was selected as the main path because:

- **it contains rich user- and computer-related semantics**
- **it supports temporally meaningful aggregation**
- **it allows before-and-after comparison through baseline correlation**
- **it can be linked more clearly to attack-related context than some of the other sources.**

For this reason, AUTH is not merely one source among others. It is the path on which the main experimental evidence of the project was built.

3.9 Methodology of the Exploratory Extensions

In addition to the main path, the project tested three exploratory extensions:

- In PROC, an End-only strategy was adopted at the advanced exploratory stage because event state appeared to be the most meaningful feature of that source.
- In FLOWS, filtering based on duration greater than zero was adopted together with aggregation appropriate for network flow behavior.
- In DNS, aggregation was based on communication pairs rather than strong filtering.

These sources were not treated as full alternatives to AUTH. Instead, they were used to test whether the pre-correlation idea could be adapted to event structures with different semantics and operational meaning.

3.10 Selection of Samples and Chunks

The project did not rely on direct processing of the entire raw dataset at once. Instead, it used samples and carefully selected chunks. This was due to two main reasons:

- a practical reason related to the size of the data and the cost of handling it in a local prototype environment
- a methodological reason related to the need for repeatable and comparable experiments before generalizing conclusions.

Therefore, using samples and chunks was not an arbitrary shortcut. It was part of a deliberate methodology that allowed before-and-after behavior to be studied through controlled units of data.

Some chunks were selected specifically from redteam-aligned regions, because the project was not interested only in numerical reduction, but also in preserving attack-related context. This led to the use of redteam-zone chunks, which made it possible to evaluate whether pre-correlation still preserved meaningful analytical signals in parts of the data that were temporally and contextually close to attack activity.

Validation across multiple chunks improved the credibility of the results, because it moved the analysis beyond a single isolated sample.

3.11 Evaluation Logic

The project was not evaluated using row counts alone. Instead, it relied on a set of connected indicators. The evaluation logic can be summarized as follows:

- stream reduction, to determine whether pre-correlation produced measurable reduction,
- change in baseline correlation outputs, to determine whether stream reduction affected downstream load,
- preservation of attack-related context, to verify that reduction did not destroy the security meaning of the stream,
- repetition across multiple chunks, to assess stability and avoid over-reliance on one sample.

This logic allowed the project to evaluate the quality of reduction, not only the quantity of reduction. This is consistent with recent work that treats alert processing and aggregation as stages that should be judged according to their analytical impact, not only their numerical compression [18], [19], [21].

3.12 Baseline Correlation Setup

Baseline correlation was used as a reference layer that allowed the project to compare the stream before and after processing. The purpose here was not to implement a complex commercial correlation engine, but to use a simple correlation mechanism that was sufficient to show the difference between the raw stream and the processed stream.

Its importance in the project lies in the fact that it:

- provided a clear reference layer for comparison
- enabled quantitative before-and-after measurement
- supported the claim that reduction was not only formal compression
- linked pre-correlation directly to a downstream analytical effect.

This means that baseline correlation was not merely a supporting step, but a central tool in demonstrating the project's practical impact [19].

3.13 Redteam-Based Validation

The project used redteam-based validation as a practical way to verify that the stream after reduction still retained signals related to attack activity. Redteam was used for:

- selecting aligned chunks
- evaluating overlap
- measuring attack-related context before and after reduction
- supporting judgment about the quality of reduction.

Methodologically, redteam was not an optional addition. It was an important validation mechanism that helped determine whether the pipeline reduced the stream without erasing signals related to attack behavior. This evaluation was also kept source-aware depending on the fields available in each source.

3.14 Output Management and Stored Results

The project did not stop at executing scripts. It also relied on organized result storage and used these stored outputs to interpret findings and make final decisions. Results were saved as summary text files and CSV comparison files, which made it possible to return to them when analyzing the performance of each strategy, comparing before-and-after behavior, and reviewing conclusions related to the main path and the exploratory extensions.

These outputs were important because they were not just execution logs. They became part of the decision logic of the project. Through them, it became possible to determine:

- which strategy should be selected as the final one
- which source should remain exploratory
- which results were the most stable and defensible
- how claims in the report could be tied to actually stored outputs.

In this sense, organized output storage became part of the work environment itself because it supported traceability and reproducibility and connected implementation with scientific writing.

3.15 Chapter Conclusion

This chapter presented the work environment and methodology used to transform the idea of temporal pre-correlation into a measurable practical implementation. It showed that the project was executed as an organized Python prototype, based on the multi-source LANL dataset, with AUTH as the main path and PROC, FLOWS, and DNS as exploratory extensions. It also clarified the tools used, the software structure of the project, the execution logic, the baseline correlation setup, the redteam-based validation method, and the way results were stored and used during analysis.

In this way, this chapter forms the bridge between the theoretical framework of Chapter Two and the proposed solution in Chapter Four, where the methodology will be translated into a concrete pipeline and the final design decisions of the project will be explained in detail.

Chapter 4: Proposed Solution

4.1 Chapter Introduction

This chapter presents the proposed solution of the project as an integrated and organized pipeline, starting from the general solution idea, moving through the working methodology and implementation logic, and ending with the main AUTH-based path and the exploratory extensions on PROC, FLOWS, and DNS. The chapter does not only describe what was built, but also explains how the solution was developed, how the final strategies were selected, and how the scientific idea was transformed into a practical pre-correlation pipeline.

The proposed solution is based on the idea that raw security streams should not be forwarded directly to correlation. Instead, they should first pass through a pre-correlation layer that reduces redundancy, removes low-value events, and aggregates temporally related events before baseline correlation [24], [25], [26]. In this way, the solution is not a replacement for correlation, but an upstream improvement layer that prepares a more meaningful stream for later analysis [24], [25].

4.2 General Architecture of the Proposed Solution

The proposed solution is built on a simple central idea: the raw security stream should be improved before it reaches baseline correlation. The purpose of this layer is to reduce repetition, remove analytically weak events, and transform temporally related events into higher-value units before they are evaluated downstream. This idea is directly supported by recent work on correlation-oriented event reduction, hierarchical correlation, and pre-correlation filtering and aggregation [24], [25], [27].

In its general form, the proposed solution consists of a sequence of connected stages. It starts with raw security logs, then moves through parsing and normalization, followed by filtering and exact de-duplication, then time-aware aggregation, baseline correlation, and finally before/after evaluation. This architecture shows that the project does not treat events as isolated raw rows only, but as a stream that can be reorganized and improved before later correlation [24], [25], [27].

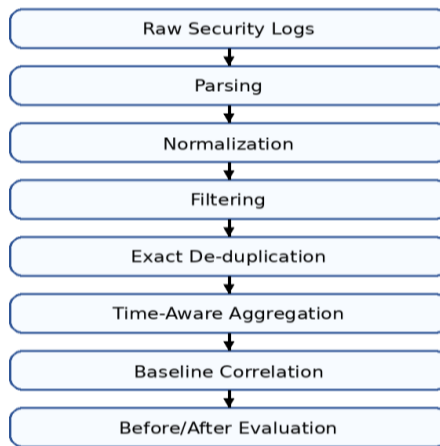


Figure 1 : General Proposed Solution

4.3 Working Methodology of the Proposed Solution

The proposed solution was not finalized in a single step. It was developed through a gradual methodology based on comparison, validation, and decision refinement. The work began by identifying the most suitable main source for implementation, then preparing the data, testing alternative reduction strategies, comparing the effect before and after baseline correlation, validating the behavior across multiple chunks, and finally selecting the most stable design.

This methodology is important because it means that the final solution was not chosen as an arbitrary first implementation. Instead, it emerged from iterative testing and repeated validation, especially in the main AUTH path. In that sense, the solution design followed a practical decision workflow: source selection, chunk preparation, alternative strategy testing, before/after comparison, validation across multiple chunks, result storage, and final strategy selection. This is also consistent with recent work that evaluates preprocessing and aggregation not only as standalone stages, but as measurable contributors to downstream analytical behavior [24], [25], [27].

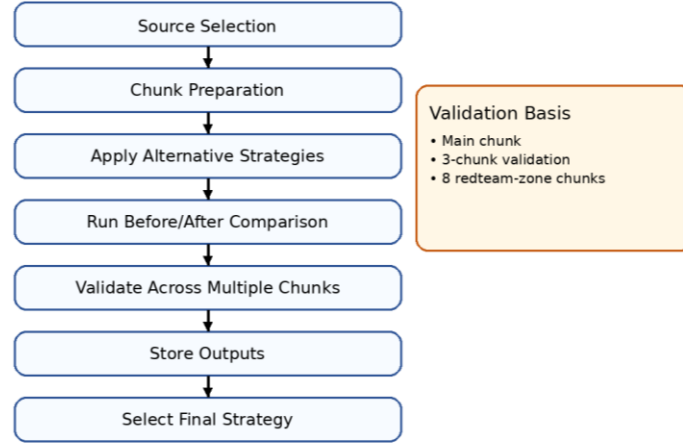


Figure 2 : Proposed Solution Implementation Mechanism

4.4 Main AUTH Path and Source-Aware Extensions

The project adopted AUTH as the main path because it was the most suitable source for building a complete end-to-end pipeline that could measure stream reduction, correlation impact, and context preservation. This path passed through parsing, normalization, improved filtering, exact de-duplication, relaxed temporal aggregation, baseline correlation, and redteam-aware evaluation. The decision to use AUTH as the main path is consistent with the idea that source-rich and semantically expressive events are the best candidates for full pre-correlation design [24], [25].

At the same time, the project did not stop at AUTH. The idea was extended to PROC, FLOWS, and DNS, but not through identical strategies. Instead, the extensions followed a source-aware logic, where each source received the strategy that best matched its event structure. PROC relied on an End-only strategy, FLOWS relied on duration > 0 with suitable aggregation, and DNS relied on source + destination aggregation. This reflects one of the core scientific positions of the project: the framework can be unified, but the implementation decisions should remain tied to source structure and meaning [25], [27], [28].

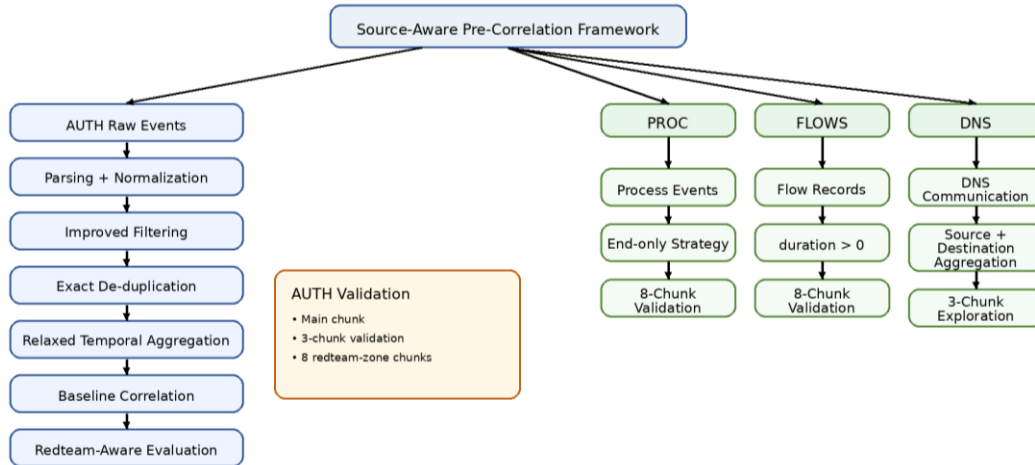


Figure 3 : Main AUTH Path and Source-Aware Extensions

4.5 Parsing and Initial Data Structuring

The parsing stage represents the practical starting point of the proposed solution. Its role is to transform raw log lines into a structured format that can be processed in later stages. In this step, the relevant fields of each event are extracted and placed into a readable tabular structure, so that the data becomes suitable for comparison, filtering, aggregation, and evaluation.

The importance of parsing is not limited to technical reading only. It also determines the structure on which the entire solution depends. Without correct parsing, the later stages cannot operate reliably. For this reason, parsing was a necessary stage across all sources, even though the exact extracted fields differed from one source to another.

4.6 Normalization

After parsing, the data passes through normalization, whose purpose is to unify the fields and structural elements needed for downstream analysis. In this project, normalization played a stronger role in the main AUTH path, because this source required clearer organization of temporal and semantic fields before filtering and aggregation.

Normalization therefore served as a structural alignment stage. It did not merely make the data cleaner; it prepared the event stream in a form that later stages could process more consistently. This was especially important in the main path, where pre-correlation depended on stable event attributes before reduction and grouping.

4.7 Improved Filtering

The improved filtering stage was designed to remove low-value or weakly useful events before deeper reduction stages were applied. The purpose of filtering was not blind deletion, but noise reduction. In other words, the goal was not simply to reduce the row count, but to improve the quality of the stream before correlation [24], [27].

In the AUTH path, improved filtering played the role of the first direct reduction layer. It decreased the size of the stream before de-duplication and aggregation, and therefore contributed to making later stages more meaningful. Its importance comes from being the first point at which the proposed solution begins to reshape the original stream instead of merely preparing it.

4.8 Exact De-duplication

After filtering, the project applied exact de-duplication. The purpose of this stage was to remove direct repetitions that did not add new analytical meaning to the stream. This stage was not intended to merge partially similar events, but to eliminate strict duplicates according to the selected event representation.

After filtering, the project applied exact de-duplication. The purpose of this stage was to remove direct repetitions that did not add new analytical meaning to the stream. This stage was not intended to merge partially similar events, but to eliminate strict duplicates according to the selected event representation.

4.9 Relaxed Temporal Aggregation

Relaxed temporal aggregation is the central stage of the proposed solution because it represents the practical use of the time element inside pre-correlation. In this stage, events are no longer treated only as separate rows. Instead, similar events that are close in time are grouped into higher-value analytical units [24], [25], [28], [29].

In the main AUTH path, this aggregation relied on a grouping key and a fixed time window, allowing the solution to combine related events without destroying their analytical meaning. It was called relaxed because it was neither too strict to prevent useful grouping nor too broad to cause unjustified merging. For this reason, this stage is the most direct practical translation of the project title into implementation, since the time element is used as an active decision factor rather than as a passive timestamp field [25], [28], [29].

4.10 Baseline Correlation Inside the Proposed Solution

After the pre-correlation stages are completed, the resulting stream is forwarded to baseline correlation so that the difference between the original and processed streams can be measured. The purpose here is not to build a large industrial correlation engine, but to use a reference layer that makes the downstream effect visible and measurable [25].

The importance of this stage is that it connects the proposed solution to later analytical outcomes. Instead of treating reduction as a simple decrease in the number of rows, the project links it to measurable changes in correlation outputs. Baseline correlation therefore became part of the proposed solution itself, not merely an external evaluation tool.

4.11 Chunk-Based Validation and Final Strategy Selection

The project did not rely on a single sample when developing the proposed solution. Instead, it used multiple chunks in order to verify that the observed behavior was not isolated. This was especially important in the AUTH path, where the work relied on a main chunk, followed by 3-chunk validation, and then 8 redteam-zone chunks for stability checking.

Through this validation process, the final strategy was chosen on the basis of repeated behavior across more than one part of the data, rather than on one isolated sample. Therefore, chunk-based validation was part of the solution development logic, not only part of the results chapter. It supported the selection of the final AUTH design and strengthened the credibility of the reduction strategy.

4.12 Exploratory Extensions for Other Sources

After stabilizing the main AUTH path, the project extended the idea to the remaining sources under a source-aware pre-correlation logic. The purpose of these extensions was not to replicate the AUTH pipeline exactly, but to test whether the same general framework could adapt to event sources with different structures and meanings [25], [27].

In PROC, the End-only strategy emerged as the most meaningful approach.

In FLOWS, the condition $\text{duration} > 0$ served as a suitable point for reducing noise before aggregation.

In DNS, aggregation over source + destination pairs was the most suitable exploratory option.

These extensions support the central idea of the project: the framework can remain unified, but the operational strategy should remain tied to the nature of the source itself [25], [27], [28].

4.13 Final Software Structure of the Solution

In its final form, the proposed solution evolved into a modular software structure instead of remaining a collection of isolated scripts. This structure included shared files for common functions, a registry for source properties, a unified runner for pipeline execution, and source-specific strategy files.

The importance of this structure lies in the fact that it made the solution more organized, easier to rerun, and clearer in terms of the relation between design logic and actual implementation. It also helped transform the project from a group of separate experiments into a more coherent execution framework.

Diagram 4. Final Modular Software Structure

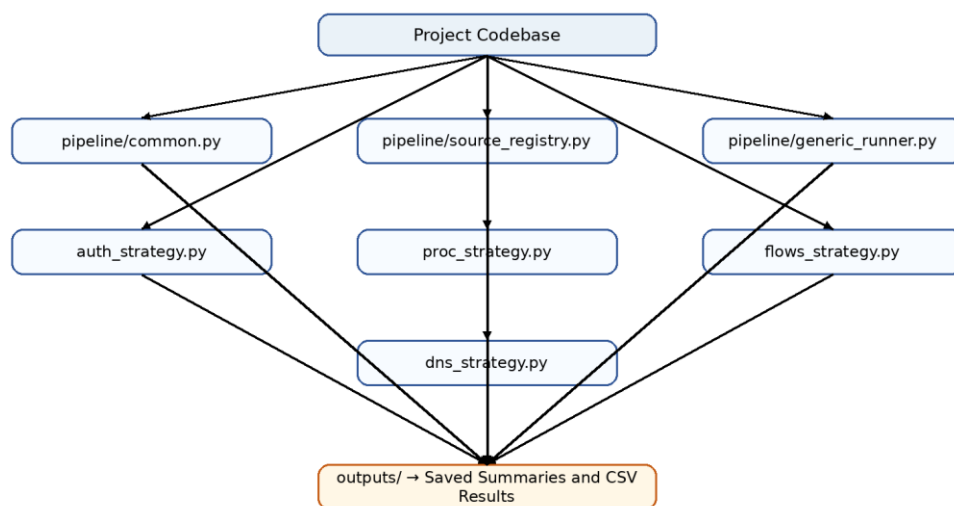


Figure 4 : Final Modular Software Structure

4.14 Final Design Decisions and Their Justifications

The final version of the proposed solution was not adopted arbitrarily. It resulted from a set of design decisions that were justified through testing and comparison. The most important of these decisions were:

- adopting AUTH as the main path,
- adopting improved filtering instead of minimal cleaning only,
- keeping exact de-duplication as an independent stage,
- adopting relaxed temporal aggregation instead of strict aggregation,
- using a fixed time window in the main path,
- selecting PROC = End-only,
- selecting FLOWS = duration > 0,
- and selecting DNS = source + destination aggregation.

This section is important because it links the final form of the solution to its methodological reasons. In other words, it explains not only what was built, but also why it was built in that way [24], [25], [27], [28].

4.15 Overall Pipeline Logic

The overall logic of the proposed solution can be summarized as follows: it begins with a raw event stream, organizes it, applies gradual reduction, forwards the result to baseline correlation, compares before and after behavior, uses stored outputs to support interpretation, and then stabilizes the final design.

This logic represents the full execution view of the solution, because it connects:

- the processing stages,
- chunk-based validation,
- source-aware extension,
- stored outputs,
- and final strategy selection.

In this sense, the proposed solution is not just a sequence of technical steps. It is a full development and execution logic for improving a SIEM-like stream before correlation.

4.16 Chapter Conclusion

This chapter presented the proposed solution as a practical pipeline for improving security event streams before baseline correlation. It showed that the solution was built on a general framework that starts from raw logs and ends with before/after evaluation, while adopting AUTH as the main path and extending the idea to PROC, FLOWS, and DNS under a source-aware logic.

The chapter also explained the detailed stages of the solution, the working methodology based on strategy testing and chunk-based validation, and the final modular structure that organized the implementation. In this way, Chapter Four provides the complete implementation view of the solution whose measurable outcomes will be presented and discussed in Chapter Five.

Chapter 5 : Comparisons and Results

5.1 Chapter Introduction

This chapter presents the project results in an organized and connected way, with a focus on the effect of the pre-correlation layer on the security event stream before baseline correlation. It does not only present numbers, but also explains their analytical meaning and shows how the reduction stages affected stream size, correlation outputs, and attack-related context preservation.

The chapter also compares the main-path results with the exploratory extensions, then discusses and interprets the results, and finally compares them with the reference studies on which the project was built. In this way, this chapter is the stage where the proposed solution moves from design and implementation methodology to interpretable and defensible results. The final project materials explicitly present the AUTH main-path results, chunk-based validation, and the cross-source comparison across AUTH, DNS, FLOWS, and PROC.

5.2 Method of Presenting the Results

The results in this project were presented using a before/after evaluation logic, meaning that the event stream was compared before and after applying pre-correlation. This is consistent with the main purpose of the project, which is to measure the effect of reduction before baseline correlation rather than only describing the stages theoretically.

Accordingly, the results are organized in this chapter in the following order:

- **results of the main AUTH path,**
- **validation results across chunks,**
- **results of the exploratory extensions,**
- **aggregated comparison across sources,**
- **discussion of results,**
- **and comparison with previous studies.**

This structure follows the same evaluation logic used in the project itself: before/after counts, redteam-related context, baseline correlation outputs, multi-chunk validation, and cross-source comparison.

5.3 Results of the Main AUTH Path

AUTH represents the main path of the project, and therefore the most complete and most important results are associated with it. This path showed that improving the stream before correlation is practically feasible and measurable, while still preserving attack-related context to a large extent. The importance of AUTH results lies in the fact that they are not only the strongest performance inside the project, but also the main evidence supporting the central project idea.

5.3.1 Stream Size Reduction

The results of the main AUTH path showed that the number of events decreased from 10000 raw events to 9439 events after filtering and relaxed temporal aggregation. This reduction indicates that pre-correlation was not a formal stage only, but achieved a real reduction in the stream before correlation. The importance of this result becomes stronger when it is interpreted together with the remaining indicators, because the project was not intended to reduce volume alone, but to reduce it without destroying the analytical meaning of the stream.

5.3.2 Effect on Baseline Correlation

The effect of the solution did not stop at reducing stream size. It also extended to baseline correlation. In the main AUTH path, the number of correlation outputs decreased from 360 to 337, which means a reduction of 23 outputs or 6.39%. This is important because it shows that reduction was not only an internal numerical change in the data, but produced a downstream effect on the later analytical stage.

This result therefore supports the main project hypothesis: improving the stream before correlation can reduce downstream burden and make the resulting outputs more focused.

5.3.3 Preservation of Attack-Related Context

In addition to reduction, the main path also showed that the redteam-related context remained close to its original level. The percentage changed only from 73.15% to 72.67%, which is a relatively small change compared with the achieved reduction and the measured change in correlation outputs. This is one of the most important findings of the project, because it shows that the solution was not blind reduction, but a measured reduction that preserved most of the attack-related signal.

Therefore, the value of the main path lies not only in reducing the stream, but in combining:

- reduction,
- measurable downstream effect,
- and near-stable preservation of context.

5.3.4 Summary of AUTH Results

To summarize the main-path findings more clearly, the following table presents the three core AUTH indicators in compact before/after form.

Table 2 : Main AUTH Path Results Summary

Metric	Before	After	Interpretation
Event stream size	10000	9439	The stream was reduced before correlation
Baseline correlation outputs	360	337	Downstream correlation load decreased
Redteam-related context	73.15%	72.67%	Attack-related context remained near-stable

The table shows that the pre-correlation layer did not only reduce the stream size, but also produced a measurable downstream effect while preserving most of the attack-related context. This is consistent with the “Main AUTH Path Results” summary in the final project presentation.

5.3.5 Visual Summary of AUTH Results

It is also useful to support these findings with one combined visual summary.

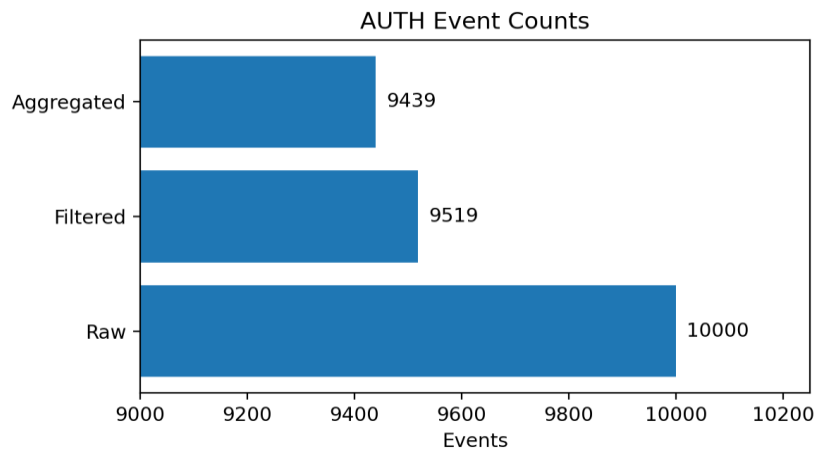


Figure 5 : compares AUTH event counts before and after processing. After filtering + relaxed temporal aggregation

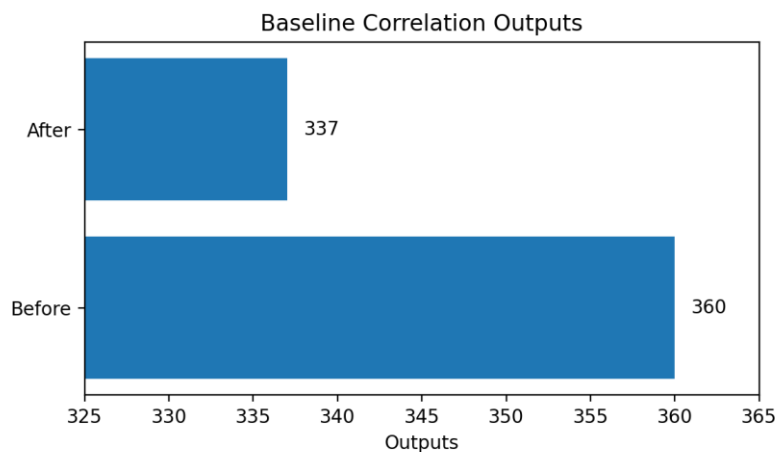


Figure 6 : : Main AUTH Path Results showing the change in correlation outputs from 360 to 337. Reduction of 23 outputs(6,3,9%)

This figure confirms that the project achieved reduction, downstream improvement, and near-stable context preservation.

The figure include:

- **Event stream: 10000 → 9439**
- **Correlation outputs: 360 → 337**
- **Redteam context: 73.15% → 72.67%**

5.4 Chunk-Based Validation Results

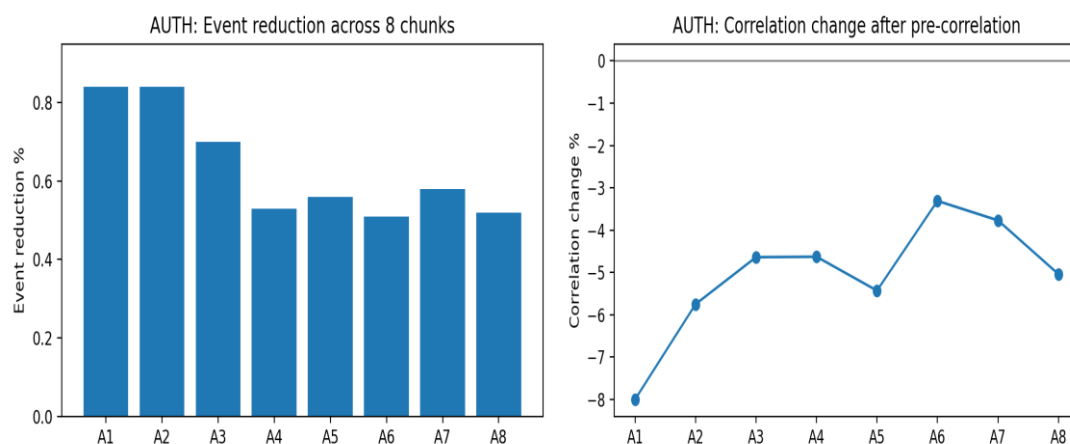
The project did not rely on a single sample when evaluating the effectiveness of the main path. Instead, it used chunk-based validation to confirm that the observed behavior was not an isolated case. This was important because any strategy may appear successful on one sample, but its scientific value becomes stronger only when it shows comparable behavior across multiple parts of the data.

5.4.1 Validation Across 3 Chunks

The project used 3-chunk validation in the main AUTH path to verify that useful reduction was not achieved only in the main chunk, but also across additional chunks. This strengthened confidence in the selected strategy because the final decision was not based on a single local observation, but on repeated behavior across multiple chunks. The final materials explicitly list AUTH validation as “3-chunk validation + 8 redteam-zone chunks.”

5.4.2 Validation Across 8 Redteam-Zone Chunks

In addition to general validation, the project used 8 redteam-zone chunks to observe the behavior of the solution in regions that are closer to attack activity. The importance of this step is that it allowed the project to examine whether reduction remained balanced even in analytically sensitive parts of the data. In this way, evaluation became stronger because it was linked not only to volume, but also to proximity to attack-related regions.



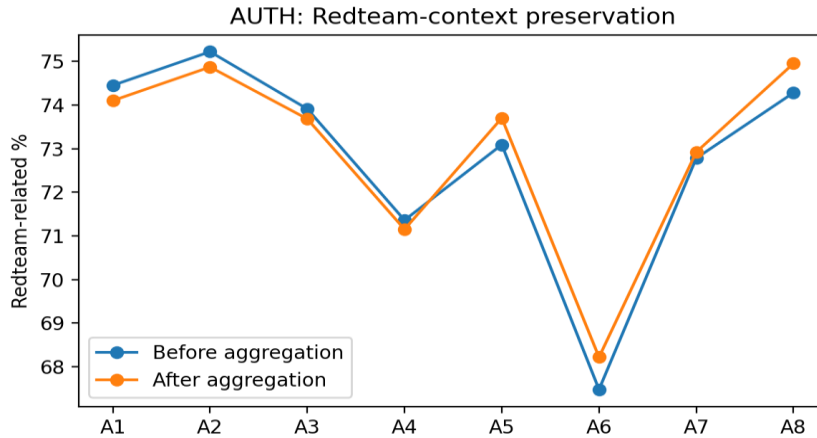


Figure 7 : AUTH Validation Across Redteam-Zone Chunks

A suitable sentence before the figure is:

To verify that the observed AUTH behavior was not limited to a single sample, the project also evaluated the selected strategy across redteam-zone chunks.

A suitable sentence after the figure is:

The figure shows that the general behavior of the strategy remained comparable across multiple chunks, which supports the stability of the final main-path design.

5.5 Results of the Exploratory Extensions

After stabilizing the main path, the project extended the idea to three additional sources: PROC, FLOWS, and DNS. The purpose of these extensions was not to compete with AUTH as the main path, but to test whether the same general idea could adapt to sources with different structures and meanings.

5.5.1 PROC Results

The results of PROC showed that the End-only strategy produced a clear improvement in concentration. The percentage increased from 10.63% to 20.65%, which made PROC one of the strongest exploratory extensions in the project. This result suggests that source-specific reduction can be more suitable than applying one uniform strategy to all sources.

5.5.2 FLOWS Results

In FLOWS, the strategy based on duration > 0 together with suitable aggregation increased the percentage from 39.14% to 42.87%. This suggests that simple filtering, when supported by appropriate aggregation, can improve concentration in flow-related events even if the event structure differs from AUTH.

5.5.3 DNS Results

In DNS, the percentage increased from 62.55% to 64.71% after using aggregation based on source + destination pairs. This suggests that DNS did not need strong filtering in the same way as some other sources, but benefited more from communication-oriented grouping.

5.6 Aggregated Comparison Across Sources

The aggregated comparison shows that the different sources did not respond in the same way, and this itself supports the main project hypothesis: implementation should remain source-aware. The final results were as follows:

- AUTH: 73.15% $\rightarrow$ 72.67%
- DNS: 62.55% $\rightarrow$ 64.71%
- FLOWS: 39.14% $\rightarrow$ 42.87%
- PROC: 10.63% $\rightarrow$ 20.65%

These values should not be interpreted superficially. A higher value does not automatically mean that a source is “better.” What matters is:

- the selected strategy,
- its effect inside that specific source,
- whether it improved concentration,
- and whether it preserved meaningful analytical signal.

For this reason, the aggregated comparison is not used only to rank the sources, but to understand how each source responded to pre-correlation.

Table 3 : Exploratory Extensions and Cross-Source View

Source	Baseline %	Aggregated %	Interpretation
AUTH	73.15%	72.67%	Main path; near-stable preservation with measurable downstream reduction
DNS	62.55%	64.71%	Exploratory source; communication-pair aggregation preserved a strong signal
FLows	39.14%	42.87%	Strong exploratory extension; duration-based reduction improved concentration
PROC	10.63%	20.65%	Strong exploratory extension; End-only sharply improved concentration

5.7 Discussion of Results

The results lead to several main conclusions. First, the main AUTH path proved that building a pre-correlation layer before baseline correlation is practically feasible and can produce a measurable downstream effect while preserving attack-related context at a near-stable level. Second, the extensions proved that the framework can remain unified, but the implementation itself must remain source-aware. Third, the comparisons showed that the success of a strategy should not be judged only by the amount of reduction, but by its ability to improve the stream without destroying its security meaning.

From this perspective, AUTH can be interpreted as the strongest main path because it is the only one that combined:

- end-to-end completeness,
- chunk-based validation,
- measurable correlation reduction,
- and context preservation.

By contrast, PROC, FLOWS, and DNS provided important value as exploratory extensions, demonstrating that the idea can be extended, but with different levels of maturity and strength.

5.7.1 Stronger Scientific Interpretation

From a scientific perspective, the results indicate that temporal pre-correlation is not only feasible, but also analytically meaningful. The project demonstrated that a pre-correlation layer can produce measurable downstream impact before baseline correlation while preserving most of the attack-related context in the main AUTH path. In addition, the exploratory extensions showed that the framework can be transferred across multiple sources, but only when implementation remains source-aware rather than blindly uniform. Therefore, the results support the central claim of the project: improving the stream before correlation is feasible, measurable, and methodologically justified. This conclusion is consistent with the final project message that the framework is unified, but the decisions must remain source-aware.

5.8 Comparison with Previous Studies

When the project results are compared with the reference studies, it becomes clear that the project is aligned with the broader direction in the literature that emphasizes event filtering, aggregation, and de-duplication before or alongside correlation. The study by Sheeraz et al. 2024 supported the inclusion of filtering, aggregation, and de-duplication in correlation-oriented workflows. The study by Maosa et al. 2024 supported the idea of transforming events into higher-level grouped units before later correlation. The study by Bajtoš et al. 2024 supported the logic of alert preprocessing and filtration in multi-step attack analysis. These methodological links are explicitly shown in Seminar 1 and in the final comparison slide.

However, the current project differs from these studies in one important respect: it did not only present pre-correlation as a general architectural idea, but measured the before/after effect directly on:

- stream size,
- baseline correlation outputs,
- redteam-related context,
- and adaptability across multiple sources. The final comparison slide also states that the project differs by measuring pre-correlation reduction and context preservation across multiple sources.

Therefore, the development introduced by this project does not lie in building a full SIEM engine or a hierarchical threat model. Instead, it lies in building a temporal pre-correlation layer that is measurable, scientifically justified, and practically tested across more than one source.

5.8.1 Comparing the Project with Reference Studies

To position the project more clearly, this section can be supported with a short comparison table.

Table 4 : Comparison with Similar Reference Studies

Criteria	Our Project	Maosa et al. 2024	Sheeraz et al. 2024
Main focus	Pre-correlation before baseline correlation	Hierarchical threat correlation	Threat correlation inside SIEM engine
Similar stages	Filtering, de-duplication, temporal aggregation, evaluation	Event aggregation before correlation	Filtering and aggregation inside a SIEM correlation workflow
Main output	Before/after measurement with context preservation	Multi-layered attack detection	Threat detection and correlation support
Distinct contribution	Direct measurement of stream reduction, correlation change, and redteam-context preservation across multiple sources	Focus on hierarchical threat modeling	Focus on SIEM engine and multi-layer attack detection

This table shows that the contribution of the project does not lie in building a full SIEM engine or a hierarchical model, but in building and measuring a temporal pre-correlation layer before baseline correlation, with practical testing across multiple sources.

5.9 Current Limitations of the Results

Although the results are important, they have clear limitations that should be stated. First, the project was implemented as a local prototype, not inside a full production SIEM environment. Second, the evaluation relied on carefully selected samples and chunks instead of running on the entire dataset at once. Third, the baseline correlation layer was used as a reference mechanism for measuring effect, not as a full industrial correlation engine. Fourth, the exploratory extensions did not reach the same maturity level as AUTH. The prototype nature of the project, the main role of AUTH, and the exploratory role of PROC, FLOWS, and DNS are all clearly stated in the project materials.

These limitations do not cancel the value of the results. Rather, they place them in their proper context: defensible academic results within an applied graduation project, and a valid basis for later development.

5.10 Chapter Conclusion

This chapter presented the project results in a before/after form and showed that the proposed solution, in the main AUTH path, achieved stream reduction, measurable reduction in baseline correlation outputs, and near-stable preservation of attack-related context. It also showed that the exploratory extensions supported the idea of source-aware pre-correlation and demonstrated that the general framework can adapt to more than one type of data.

By comparing the results with previous studies, it became clear that the project is consistent with recent directions in pre-correlation and alert reduction, while offering a clear applied contribution: direct measurement of before/after effect and explicit linking of reduction to downstream attack-related context. In this way, this chapter prepares the natural ground for the general conclusion of the report, where the final contribution, limitations, and future directions of the project are summarized.

Conclusion

This project addressed the topic of Enhancing SIEM systems using time correlation by designing and implementing a temporal pre-correlation layer before baseline correlation. The purpose of this layer was to improve the security event stream before downstream analysis instead of relying only on correlation after raw events had already entered the pipeline. The project started from a clear scientific problem: modern SIEM environments ingest large volumes of heterogeneous, time-stamped alerts and logs, and these raw streams often contain redundancy, noise, and fragmented evidence, which reduce analytical clarity and increase the burden on downstream correlation outputs.

Based on this problem, the project was developed as a Python prototype using the LANL Cyber Security Dataset, with AUTH as the main path and PROC, FLOWS, and DNS as exploratory extensions under a source-aware pre-correlation framework. The proposed solution was built as a sequence of connected stages including parsing, normalization, improved filtering, exact de-duplication, relaxed temporal aggregation, baseline correlation, and before/after evaluation.

The results of the main AUTH path showed that the project was able to reduce the stream size from 10000 events to 9439, while also reducing baseline correlation outputs from 360 to 337, which represents a decrease of 23 outputs or 6.39%. At the same time, the redteam-related context remained close to its original level, changing only from 73.15% to 72.67%. These results indicate that the project did not achieve only numerical reduction, but also produced a measurable downstream effect while preserving most of the attack-related context.

The exploratory extensions also supported the central project idea. They showed that the framework can remain unified, but the applied strategies must still depend on the nature of each source. In PROC, the End-only strategy proved meaningful. In FLOWS, the condition $\text{duration} > 0$ served as a useful reduction point. In DNS, source + destination aggregation was the most suitable exploratory option. This confirms that source-aware processing was not a secondary detail, but one of the main methodological conclusions of the project.

When the results were compared with previous studies, it became clear that the project is consistent with recent directions that emphasize filtering, aggregation, and de-duplication before or alongside correlation. However, the project adds a clear applied contribution by measuring the before/after effect directly, and by linking reduction not only to stream size, but also to baseline correlation outputs and attack-related context preservation across multiple sources.

Despite these findings, the project still has clear limitations. It was implemented as a local prototype, the evaluation relied on selected samples and chunks rather than the entire dataset at once, and the baseline correlation stage served as a reference layer rather than a full industrial SIEM engine. In addition, the exploratory extensions did not all reach the same level of maturity as the AUTH path. Nevertheless, these limitations do not reduce the value of the project; rather, they define its correct scope as an applied graduation project that provides a valid basis for future development.

In light of this, several future directions can be proposed, such as connecting the framework to a more realistic correlation engine, testing it on additional datasets, studying adaptive temporal windows, extending it toward a more realistic streaming architecture, and considering machine learning later only after stabilizing the current interpretable pre-correlation layer.

Overall, the project concludes that improving the stream before correlation is feasible, measurable, and scientifically justified, and that time in pre-correlation should not be treated only as a descriptive field, but as an active analytical factor that can reduce noise, improve representation, and preserve security meaning before correlation.

References

References for Chapter One

[1] Sheeraz, M., et al. Revolutionizing SIEM Security: An Innovative Correlation Engine Design for Multi-Layered Attack Detection. Sensors, 2024.

<https://www.mdpi.com/1424-8220/24/15/4901>

[2] Maosa, H., Ouazzane, K., & Ghanem, M. C. A Hierarchical Security Event Correlation Model for Real-Time Threat Detection and Response. Network, 2024.

<https://www.mdpi.com/2673-8732/4/1/4>

[3] Ban, T., Takahashi, T., Ndichu, S., & Inoue, D. Breaking Alert Fatigue: AI-Assisted SIEM Framework for Effective Incident Response. Applied Sciences, 2023.

<https://www.mdpi.com/2076-3417/13/11/6610>

[4] Bajtoš, T., Sokol, P., & Kurimský, F. Processing of IDS Alerts in Multi-Step Attacks. Software Impacts, 2024.

<https://www.sciencedirect.com/science/article/pii/S2665963824000101>

[5] Roelofs, T.-M., et al. Finding Harmony in the Noise: Blending Security Alerts for Attack Detection. 2024.

<https://doi.org/10.1145/3605098.3635981>

[6] Zhang, R., et al. Intrusion Alert Analysis Method for Power Information Communication Networks Based on Data Processing Units. Information, 2025.

<https://www.mdpi.com/2078-2489/16/7/547>

[7] Špaček, S., et al. HTTPS Event-Flow Correlation: Improving Situational Awareness in Encrypted Web Traffic. 2022.

<https://doi.org/10.1109/NOMS54207.2022.9789877>

[8] Werner, G., & Yang, S. J. Near Real-Time Intrusion Alert Aggregation Using Concept-Learning. 2021.

<https://doi.org/10.1145/3457388.3458663>

References for Chapter Two

[9] Sheeraz, M., et al. Revolutionizing SIEM Security: An Innovative Correlation Engine Design for Multi-Layered Attack Detection. Sensors, 2024.

<https://www.mdpi.com/1424-8220/24/15/4901>

[10] Maosa, H., Ouazzane, K., & Ghanem, M. C. A Hierarchical Security Event Correlation Model for Real-Time Threat Detection and Response. Network, 2024.

<https://www.mdpi.com/2673-8732/4/1/4>

[11] Ban, T., Takahashi, T., Ndichu, S., & Inoue, D. Breaking Alert Fatigue: AI-Assisted SIEM Framework for Effective Incident Response. Applied Sciences, 2023.

<https://www.mdpi.com/2076-3417/13/11/6610>

[12] Bajtoš, T., Sokol, P., & Kurimský, F. Processing of IDS Alerts in Multi-Step Attacks. Software Impacts, 2024.

<https://www.sciencedirect.com/science/article/pii/S2665963824000101>

[13] Roelofs, T.-M., et al. Finding Harmony in the Noise: Blending Security Alerts for Attack Detection. 2024.

<https://doi.org/10.1145/3605098.3635981>

[14] Zhang, R., et al. Intrusion Alert Analysis Method for Power Information Communication Networks Based on Data Processing Units. Information, 2025.

<https://www.mdpi.com/2078-2489/16/7/547>

[15] Špaček, S., et al. HTTPS Event-Flow Correlation: Improving Situational Awareness in Encrypted Web Traffic. 2022.

<https://doi.org/10.1109/NOMS54207.2022.9789877>

[16] Werner, G., & Yang, S. J. Near Real-Time Intrusion Alert Aggregation Using Concept-Learning. 2021.

<https://doi.org/10.1145/3457388.3458663>

[17] Larroche, C., et al. Node and Edge Role Embeddings for Atypical Behavior Detection in Cybersecurity Data. 2022.

<https://arxiv.org/abs/2103.15708>

References for Chapter Three

[18] Sheeraz, M., et al. Revolutionizing SIEM Security. Sensors, 2024.

<https://www.mdpi.com/1424-8220/24/15/4901>

[19] Maosa, H., et al. A Hierarchical Security Event Correlation Model. Network, 2024.

<https://www.mdpi.com/2673-8732/4/1/4>

[20] Ban, T., et al. Breaking Alert Fatigue. Applied Sciences, 2023.

<https://www.mdpi.com/2076-3417/13/11/6610>

[21] Bajtoš, T., et al. Processing of IDS Alerts in Multi-Step Attacks. Software Impacts, 2024.

<https://www.sciencedirect.com/science/article/pii/S2665963824000101>

[22] Larroche, C., et al. Node and Edge Role Embeddings for Cybersecurity. 2022.

<https://arxiv.org/abs/2103.15708>

[23] Kent, Alexander D. Comprehensive, Multi-Source Cyber-Security Events Data Set. 2015. DOI: 10.17021/1179829.

<https://www.osti.gov/biblio/1179829>

References for Chapter Four

[24] Sheeraz, M., et al. Revolutionizing SIEM Security: An Innovative Correlation Engine Design for Multi-Layered Attack Detection. *Sensors*, 2024.

<https://www.mdpi.com/1424-8220/24/15/4901>

[25] Maosa, H., Ouazzane, K., and Ghanem, M. C. A Hierarchical Security Event Correlation Model for Real-Time Threat Detection and Response. *Network*, 2024.

<https://www.mdpi.com/2673-8732/4/1/4>

[26] Ban, T., Takahashi, T., Ndichu, S., and Inoue, D. Breaking Alert Fatigue: AI-Assisted SIEM Framework for Effective Incident Response. *Applied Sciences*, 2023.

<https://www.mdpi.com/2076-3417/13/11/6610>

[27] Bajtoš, T., Sokol, P., and Kurimský, F. Processing of IDS Alerts in Multi-Step Attacks. *Software Impacts*, 2024.

<https://www.sciencedirect.com/science/article/pii/S2665963824000101>

[28] Werner, G., and Yang, S. J. Near Real-Time Intrusion Alert Aggregation Using Concept-Learning. 2021.

<https://doi.org/10.1145/3457388.3458663>

[29] Špaček, S., et al. HTTPS Event-Flow Correlation: Improving Situational Awareness in Encrypted Web Traffic. 2022.

<https://doi.org/10.1109/NOMS54207.2022.9789877>